

CAREER: Foundations of Cost-Aware Energy Characterization and Optimization for Computing Systems

Overview

Modern computing is entering an energy-constrained era. AI accelerators and HPC systems are rapidly increasing electricity demand, operating cost, and pressure on regional energy resources. Yet research in computer systems and architecture still treats energy mostly as a scalar quantity: joules, power, or electricity price. This abstraction is increasingly inadequate. The same unit of energy can carry different cost implications depending on when and where it is consumed, how it is generated, and what hardware lifecycle stage it supports. Systems that appear energy-efficient may therefore remain cost-inefficient in practice.

This CAREER project will establish the foundations of cost-aware energy characterization and optimization for computing systems. The central theme is that computing systems must reason about the cost-relevant properties of energy, not only the amount of energy consumed. The project has three thrusts: define cost-aware energy metrics and models, measure hidden energy costs across AI and HPC systems, and optimize hardware and runtime decisions using these new energy-cost signals. Together, these thrusts will make energy cost a first-class systems abstraction: something computing systems can quantify, attribute, predict, and optimize across hardware and software decisions.

Keywords: Cost-aware energy modeling; energy-efficient computing; lifecycle modeling

Intellectual Merit

This project advances computer systems and architecture by developing a new foundation for treating energy cost as a first-class systems abstraction. Existing systems research has made major progress in reducing energy consumption, but it often treats energy as a uniform quantity. This project challenges that abstraction by modeling energy according to its cost-relevant properties, including workload behavior, hardware design, lifecycle stage, deployment context, and time-varying operating conditions.

The research will contribute new abstractions, models, and algorithms for cost-aware energy characterization and optimization. First, it will define metrics that connect computing activities to energy-cost drivers across hardware and software decisions. Second, it will develop measurement methods to identify where hidden energy costs arise in AI and HPC systems. Third, it will design optimization techniques for architecture design space exploration, workload scheduling, and runtime adaptation. These contributions will expose tradeoffs that are invisible to conventional energy metrics and enable new Pareto-optimal designs that reduce real cost while maintaining performance and scalability.

Broader Impacts

This project will advance U.S. leadership in energy-efficient AI infrastructure and semiconductor innovation by helping computing systems reduce the real cost of energy use. The results can inform semiconductor design, HPC operation, hardware procurement, workload management, and regional deployment decisions, enabling more cost-efficient and scalable computing infrastructure. The project will also advance education and workforce development. New course modules, open-source tools, datasets, and hands-on projects will train students in cost-aware energy modeling and hardware-software optimization. Outreach through Purdue programs will introduce K–12 and undergraduate students to how computer systems and architecture design can shape the future of energy-efficient, scalable computing infrastructure.